

Yazdan Zandiye Vakili

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🌐 [LinkedIn Profile](#)

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Education

University of Tehran

Tehran, Iran

Bachelor of Computer Science

September 2018 - Jan 2024

- Overall GPA: **17.6/20 (3.7/4)**

National Organization for Development of Exceptional Talents

Tehran, Iran

Diploma in Mathematics and Physics Discipline

September 2014 - June 2018

- Overall GPA: **19.51/20 (4/4)**

Research Interests

- Natural Language Processing
- Information Retrieval
- Software Engineering
- Security & Privacy
- Applied Machine Learning
- Image Processing

Work Experience

- **Workflow Company GmbH** | Software Engineer (Junior) | Supervisors : [Gabriel Gruber](#) , [Helga Gruber](#) **Aug 2025 - Present**

- **Workflow Company GmbH** | Software Engineer ([IAESTE](#) Internship) | Supervisors : [Gabriel Gruber](#) , [Helga Gruber](#) **Sep 2024 - Aug 2025**

[Vienna, Austria](#)

- **Java** (Spring Boot framework, Vaadin, Hibernate6, Junit5)
- **JavaScript** (React Framework)
- Git (Version control)
- MySQL & SQLServer (Database)
- **Docker & Kubernetes**

- **University of Tehran** | Teaching Assistant **Jan 2020 - Jan 2025**

Tehran, Iran

- Teaching assistant at Artificial Intelligence course supervised by [Dr. Hedieh Sajedi](#)
- Teaching assistant at Data Mining course supervised by [Dr. Hedieh Sajedi](#)
- Teaching assistant at Image Processing (Graduate Course) supervised by [Dr. Hedieh Sajedi](#)
- Teaching assistant at Advanced Information Retrieval supervised by [Dr. Bagher BabaAli](#)
- Teaching assistant at 8 other courses too
- Teaching assistant at Introduction to Operating Systems supervised by [Dr. Hedieh Sajedi](#)
- Teaching assistant at General Mathematics II supervised by [Dr. Reza Rokni](#)
- Teaching assistant at Differential Equations supervised by [Dr. Reza Rokni](#)
- Teaching assistant at Theory of Computation supervised by [Dr. Fatemeh Halataei](#)
- Teaching assistant at Basic Programming (2 semesters) supervised by [Dr. Marjan Goodarzi](#)
- Teaching assistant at Basic Programming supervised by [Dr. Mahdi Seyfipoor](#)
- Teaching assistant at Machine & Assembly Language supervised by Dr. Rafei
- Teaching assistant at Introduction to Bioinformatics supervised by [Dr. Kaveh Kavousi](#)

- **Karyar College** | Python, Django & Data Analysis Tutor & Course Supervisor | Supervisors : [Negar Rahmati](#) , **Jul 2020 - Oct 2024**

[Niloufar fazlollahi](#) , [Dr. Arezou Keshavarz](#)

Tehran, Iran

- Django framework **course supervisor**
- Advanced Python **course supervisor**
- Advanced Python Tutor(**Socket Programming, MultiThreading, OOP**)
- Basic data analysis Tutor(**Pandas, Numpy, Matplotlib, SQL**)
- Entrance Workshop Tutor (Basic Algorithms)
- Algorithm Workshop Tutor (**Algorithm Design, Data Structure, Linear Algebra, Introduction to Probability**)
- Three semester as Advanced Python Teaching Assistant
- Two semesters as Backend Development with Django Teaching Assistant (**REST Framework, ORM, Postgre, SQLite**)

- **Tadbir Pardaz Company** | Part-time Backend Developer **Dec 2022 - Dec 2023**

Tehran, Iran

- **Launching Confluence System** for documentations in the company & Documenting codes
- Backend Development with **ASP.Net, C#**
- Handle the entrance of the customers data on database (**T-SQL**)

- Working in a **Scrum** based environment and got familiar with the concepts
- Learned how to handle the task (**Azure DevOps**) and tune myself with company's flow
- Worked and got familiar with **RabbitMQ** and **Design Patterns** and **Microservice** system design
- Got more experienced with **Git** and **project management** systems

• **TaarLab Laboratory** | Part-time AI Research Assistant | Supervisor : [Dr. Taleh Masouleh](#)

May 2023 - Sep 2023

Tehran, Iran

- Use Deep Learning algorithms for **Image segmentation** and **Image detection**
- Design the best ANN with good accuracy on detecting the grasp points on selected items in the dataset
- Using tools like **Numpy**, **Pandas**, **Mediapip**, **Tensorflow**, **Pytorch**, **cv2**, **matplotlib**, **seaborn**, **transformers**, **sci-kit learn** ...

Publications

(Citations: 25 | h-index: 2)

• **Distilled BERT Model In Natural Language Processing** | [Paper](#) | Citations: 13

February 2025

Supervisor: [Dr. Hedieh Sajedi](#)

This paper reviews the evolution of NLP models, focusing on distillation techniques for creating smaller, efficient versions of large models. Key methods like TinyBERT, DistilBERT, and MobileBERT are analyzed for their ability to maintain performance while reducing size, enabling deployment on resource-limited devices. **The analysis and comparisons were shaped by my contributions to the study.**

• **Enhancing Sentiment Analysis of Persian Tweets: A Transformer-Based Approach** | [Paper](#) | Citations: 10

May 2024

Supervisor: [Dr. Hedieh Sajedi](#)

This paper improves sentiment analysis of Persian Twitter by combining models. **I initiated and implemented the hybrid approach**, integrating Naive Bayes, a rule-based method, and BERT, achieving 89% accuracy, surpassing BERT's 86%. Despite added complexity, it matches BERT's efficiency, enhancing social media sentiment analysis.

• **AI-driven approaches for dysgraphia diagnosis using online and offline handwriting data: A comprehensive scoping review** | [Paper](#) | Citations: 1

December 2025

Supervisor: [Dr. Hedieh Sajedi](#)

This paper reviews AI models, such as CNNs and SVMs, for diagnosing dysgraphia through handwriting analysis, achieving over 90% accuracy despite challenges like small datasets and language dependence. **I contributed by evaluating these models and recommending future research with larger, language-independent datasets.**

• **Revolutionizing Dysgraphia Detection: Combining Feature Fusion with Non-Discriminatory Regularization** | [Paper](#) | Citations: 1

May 2025

Supervisor: [Dr. Hedieh Sajedi](#)

AI-Based Dysgraphia Detection Using Feature Fusion and Non-Discriminatory Regularization This study presents a novel deep learning framework for diagnosing dysgraphia in children by integrating feature fusion and non-discriminatory regularization techniques. The proposed model, which replaces soft voting with a 5-layer neural network, achieves 99.68% accuracy and is optimized for real-time, web-based diagnostics. It offers a scalable, bias-reduced solution for educational and healthcare applications, with potential for personalized intervention and longitudinal tracking.

• **An Approach to Accurate Recognition of Emotions through Speech-to-Image Signal Conversion and Deep Convolutional Neural Networks** | [Paper](#)

August 2025

Supervisors: [Ajith Abraham](#), [Arash Ahmadi](#), [Dr. Mohammad Reza Falahzadeh](#)

This paper proposes a chaogram-based method for speech emotion recognition using DCNNs, achieving up to 97.68% accuracy on EMO-DB and 92.34% on eINTERFACE05. **I developed the chaogram approach and model framework, demonstrating its effectiveness .**

• **A comprehensive comparison of various drug synergy score prediction methods** | [Paper](#)

January 2024

Supervisor: [Dr. Changiz Eslahchi](#)

In this paper, **I led the analysis of modern techniques for predicting drug synergy in cancer treatments**, evaluating their strengths and limitations to advance precision oncology and guide targeted therapies.

• **Cross-Corpus Speech Emotion Recognition Using a Three-Dimensional Convolutional Neural Network with Hyperparameters Tuned by Gray Wolf Optimizer.** | Submitted (IEEE Access)

October 2025

Supervisors: [Ajith Abraham](#), [Dr. Mohammad Reza Falahzadeh](#)

Proposed a deep learning method for speech emotion recognition using **Reconstructed Phase Space (RPS)** features and a 3D CNN optimized via the **Gray Wolf Optimization** algorithm. The approach adaptively selects time lag parameters to improve emotion-specific representation and achieves state-of-the-art cross-corpus accuracy on benchmark datasets (Berlin EMO-DB and eINTERFACE'05).

- **Combining Minimum Spanning Tree and Label Propagation for Robust Pathway Enrichment: A Case Study on Glioblastoma and Colorectal Cancer** | Submitted (Molecular Medicine)

August 2025

Supervisor: [Dr. Kaveh Kavousi](#)

This paper presents a novel gene enrichment method for GBM and COAD, combining MST and LPA. Using TCGA data and STRING-based PPI networks, it identified key interactions like BEST3-METTL1 in GBM and AREG-AXIN2 in COAD, revealing cancer pathways. **I developed this method, offering insights into molecular mechanisms and therapies.**

Honors and Awards

- Faculty of Engineering Top Student of the Year (the FOE Award) | [View Certificate](#)

September 2019

Programming Skills

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|--|--|--------------------|
| • Python (Tensorflow, Numpy, Pandas, Django) | • C# & C++ (Object-Oriented Programming) | • SQL |
| • Java (Spring Boot, Hibernate6, Vaadin, Jnuit5) | • JavaScript (React Framework) | • HTML & CSS |
| • Docker & Kubernetes | • Redis | • RabbitMQ & Kafka |

Languages

- Persian (Farsi) | Native
- English | Advanced (**IELTS** - Overall score: **8** | [Results](#))
- German (Duetsch) | Elementry (A2)